



Sustainability Track : Carbon Footprint of LLMs

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Team 15



SUMMARY

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- SOLUTION
- DATA PREPROCESSING
- CLEAN DATASET
- MODEL CREATION
- CARBON FOOTPRINT ANALYSIS
- GLOBAL ARCHITECTURE
- CHALLENGES FACED
- DEMO
- CONCLUSION & FUTURE IMPROVEMENTS

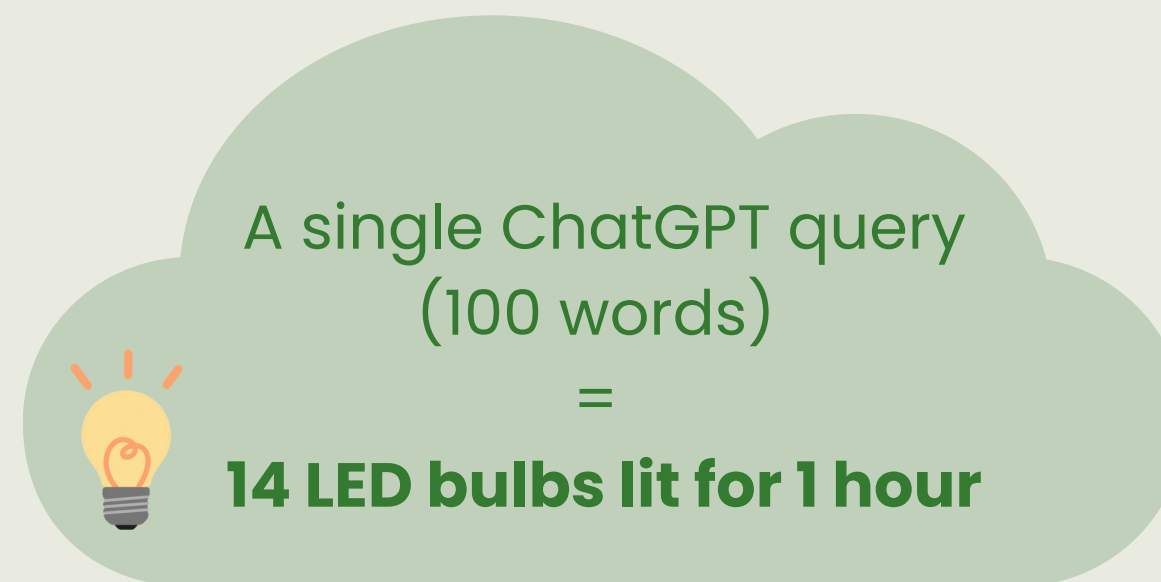
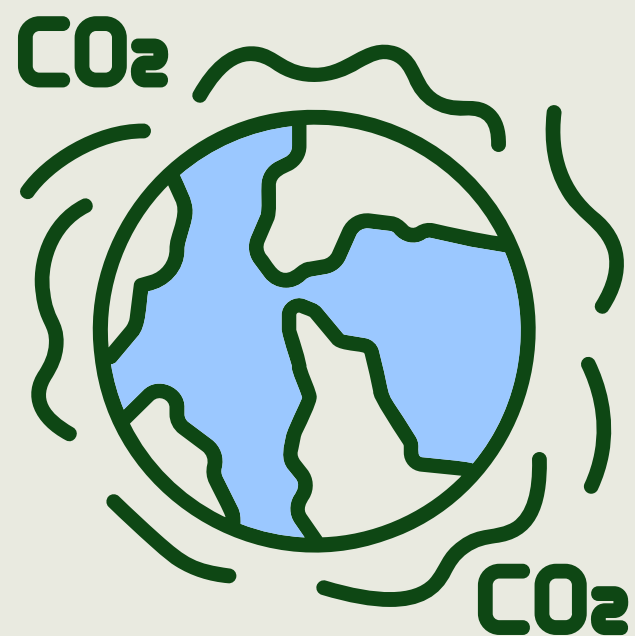
THE PROBLEM

What is the CO₂ impact of your LLM queries?

**LLM models consume
a lot of energy.**

**Users have no
visibility into their
carbon footprint.**

**No simple solution
exists to measure or
visualize this impact
in real time.**



According to a researcher at the University of Riverside in the United States: [Paper](#)

SOLUTION

A google extention showing you the impact of your queries



Setup your current localisation

France ▼

Submit

Query Dashboard

Type your query

Qu'est ce que le prompt engineering

Send

Query Dashboard

< Back

Your query

Qu'est ce que le prompt engineering?

Selected country

France

Response from Llama

Le prompt engineering est un concept qui a émergé récemment dans la communauté des modèles de langage artificiels, notamment avec l'apparition d'intelligence artificielle (IA) et de réseaux de neurones. En anglais, le terme "prompt engineering" désigne la technique de créer des instructions spécifiques ou des stimuli pour un modèle de langage artificiel afin de stimuler sa capacité à générer du contenu pertinent, précis et utile. Ces instructions sont appelées "prompts" (en français : "encadrages") et servent à guider le modèle dans son processus de traitement du langage. Le prompt engineering repose sur plusieurs principes : 1. ****Connaissance de la modélisation**** : Le responsable doit connaître les caractéristiques

Query Dashboard

This week

Nice chart

5 g of CO2



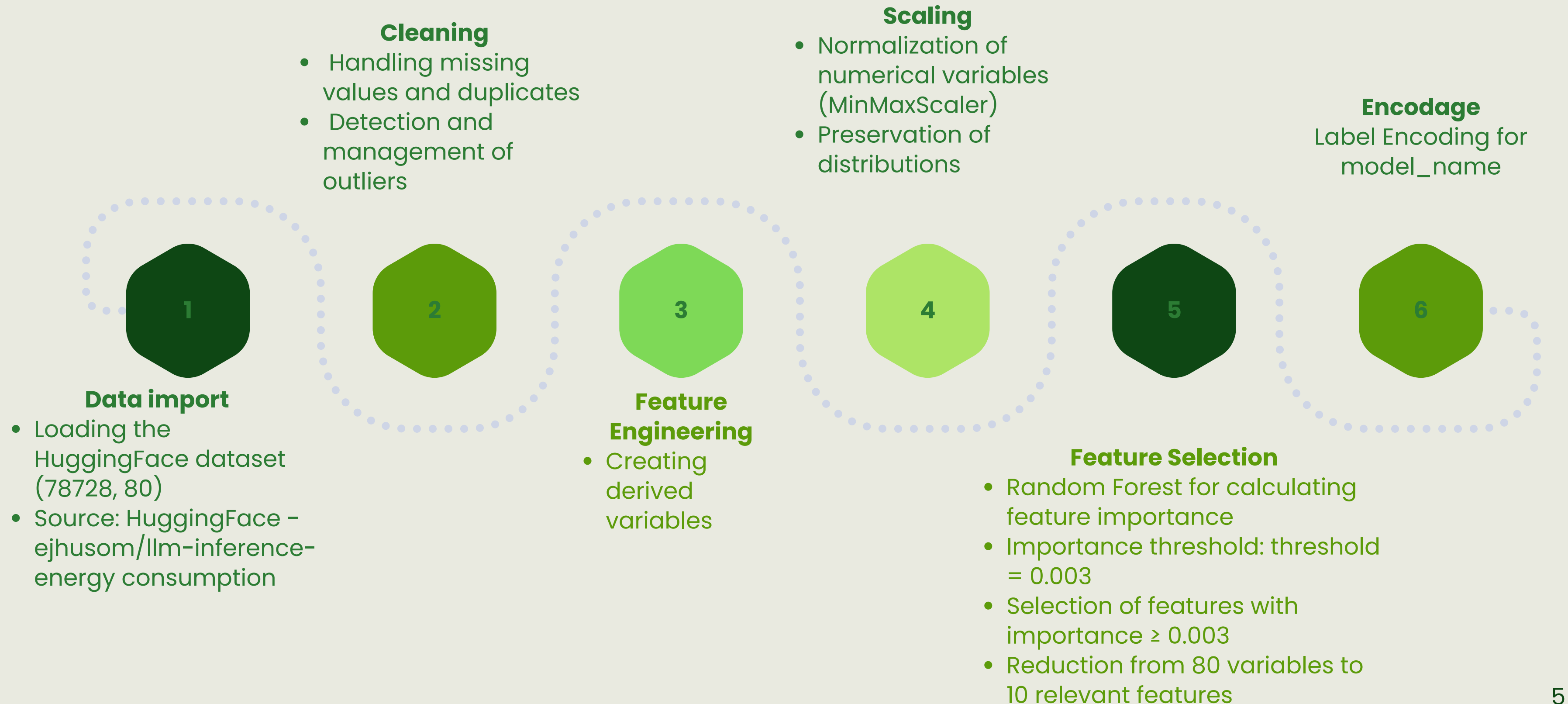
Your consommation weekly



42 Google Research

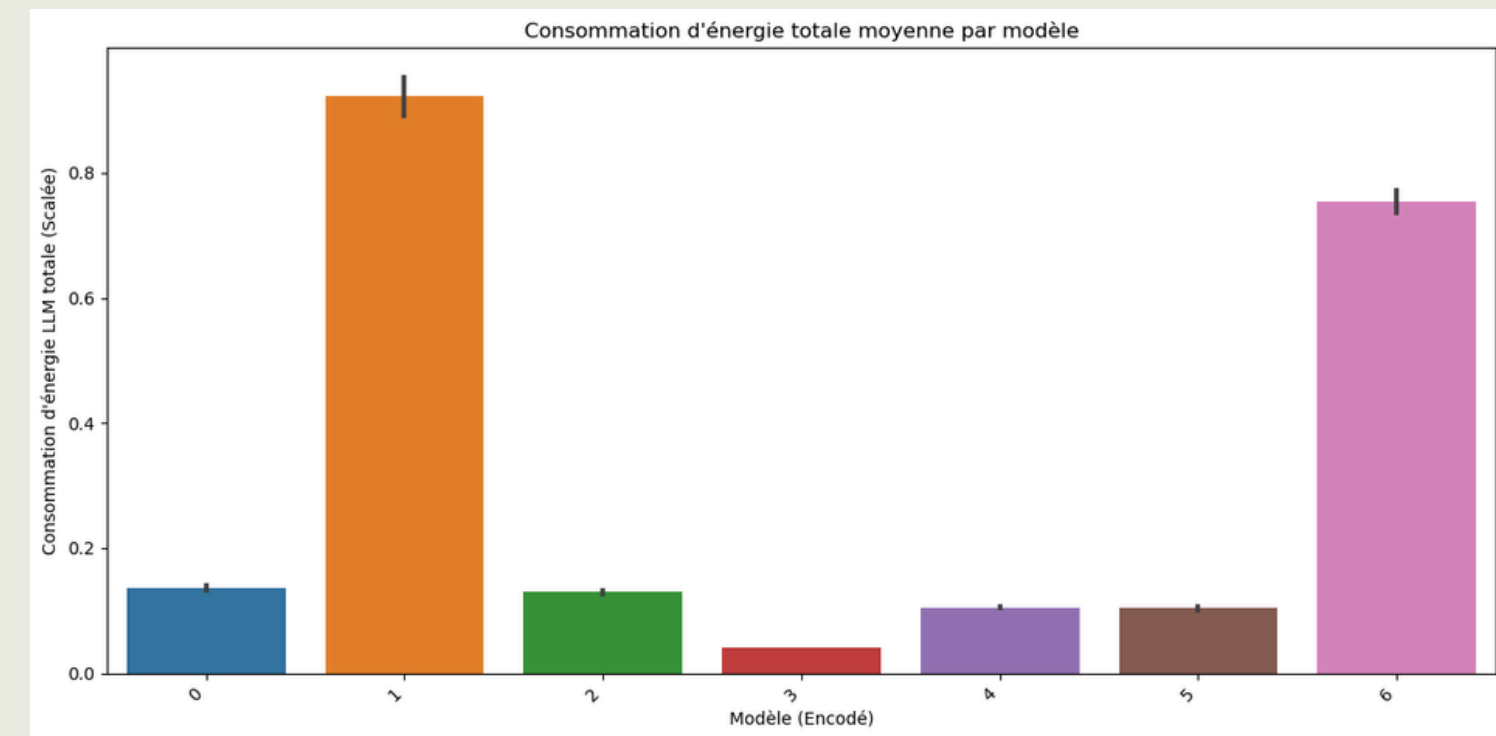
70 hectares of forest

DATA PREPROCESSING

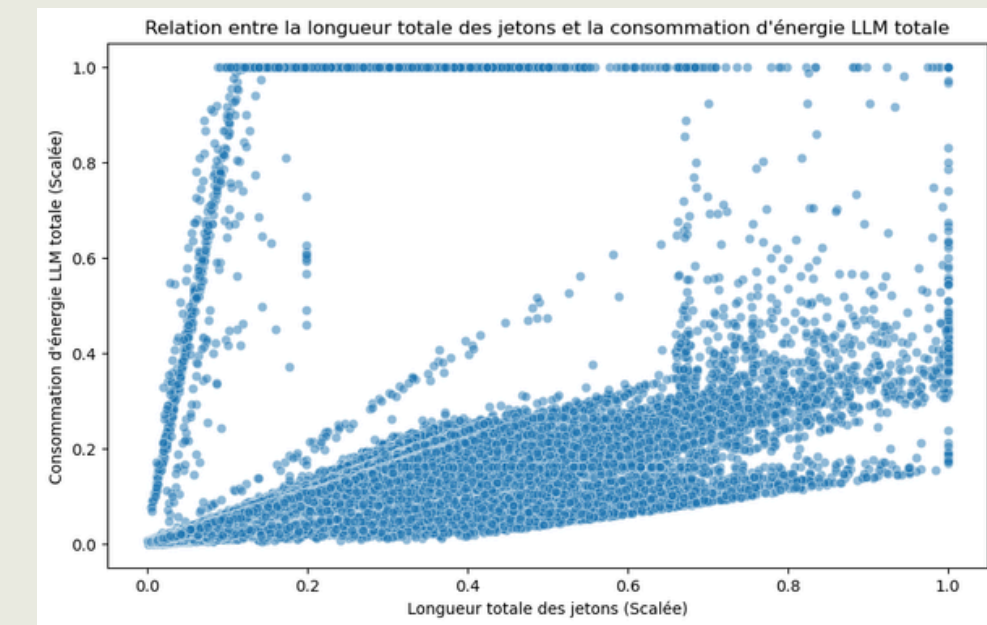
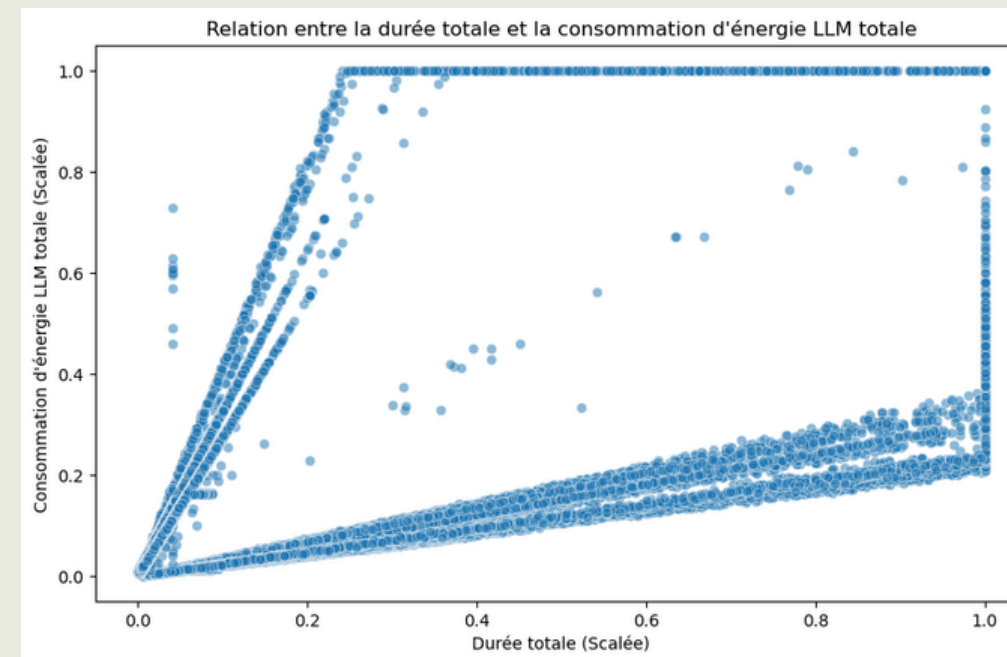


CLEAN DATASET

- **Number of rows:** 73377
- **Number of features :** 10
 - Response_token_length
 - Total_token_length
 - Response_token_length
 - Total_inference_duration
 - Response_speed_tps
 - Prompt_duration
 - Prompt_speed_tps
 - Response_duration
 - Total_duration
 - model_name_encoded
 - Energy_consumption_llm_total
- **TARGET:**
Energy_consumption_llm_total



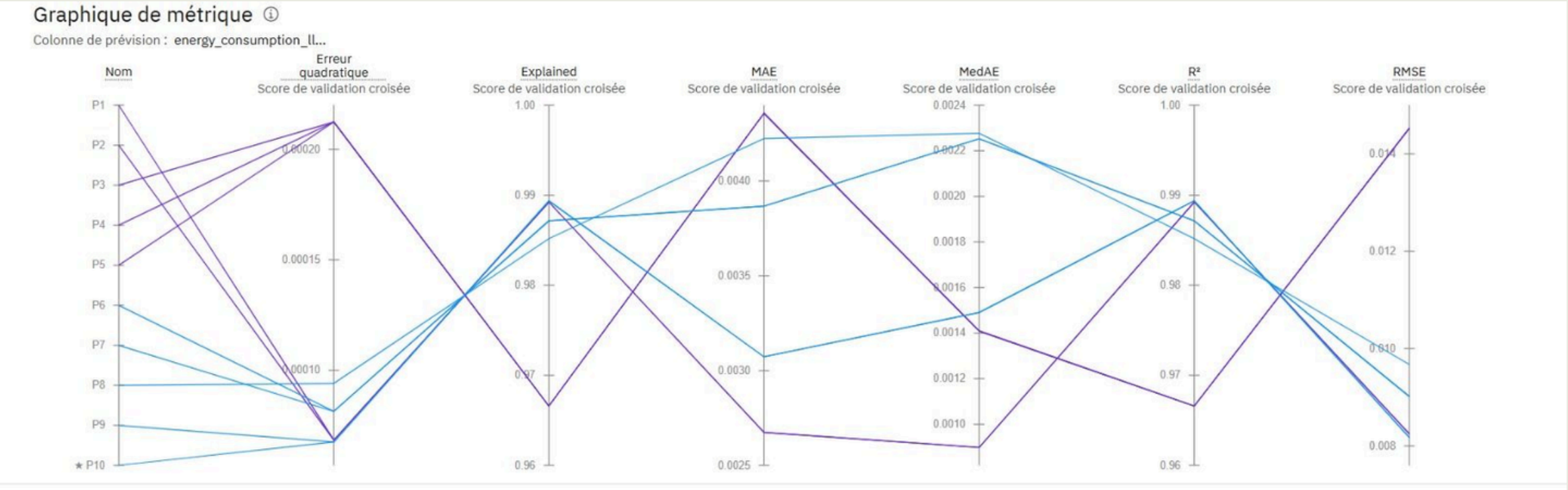
codellama: 0
codellama:7
0b: 1
codellama:7
b: 2
gemma:2b: 3
gemma:7b: 4
llama3': 5
llama3:70b: 6



MODEL CREATION

Model Training

- Used IBM Watsonx AutoAI to test and optimize multiple regression models.
- Selected Random Forest Regressor as the best performer (high R^2 , low MAE/RMSE)
- Automatic comparison of multiple algorithms



Evaluation de l'entraînement

Métrique	Données de formation	Données d'exclusion
Explained variance	0.98937434	0.98578936
Mean absolute error	0.0030725645	0.0030243846
Mean squared error	0.00006757313	0.00009092308
Mean squared log error	0.000053106774	
Median absolute error	0.0014904797	0.0014054556
R squared	0.98937416	0.9857887
Root mean squared error	0.008173015	0.009535359
Root mean squared log e rror	0.0072874394	

Deployment

- We deploy the model with IBM watsonx
- The model is now accessible with an API and a token to predict the energy_consumption of a prompt

CARBON FOOTPRINT ANALYSIS

Total emissions (kg CO₂e) :

$$CO_2e = energy_consumption_llm_total \times PUE \times emission_factor$$

- PUE: 1.05 (local)
- emission factor France: 0.051 kg CO₂e/kWh

Forest area :

$$A_acres = (E_CO_2e \times 31,536,000) / (1000 \times t)$$

- E_CO₂e = total emissions (kg CO₂e)
- t = total duration (seconds)
- ≈ 1 acre captures ~1 ton CO₂ each year

Google searches :

$$N_searches = E_CO_2e / 0.00123$$

- 1 search = 1.23 g CO₂e

 **10,000 kg CO₂e emitted in France**

 **≈ 63 acres of forest to offset**

 **≈ 8.1 million Google searches**

GLOBAL ARCHITECTURE

Chrome Extension

Interface + input + display

Send request

Backend

Central orchestrator

LLama API

Answer request
Return metric

Send metrics

ML model

Input : metrics
Output : energy
consumption

Send energy consumption

Calculation Modules

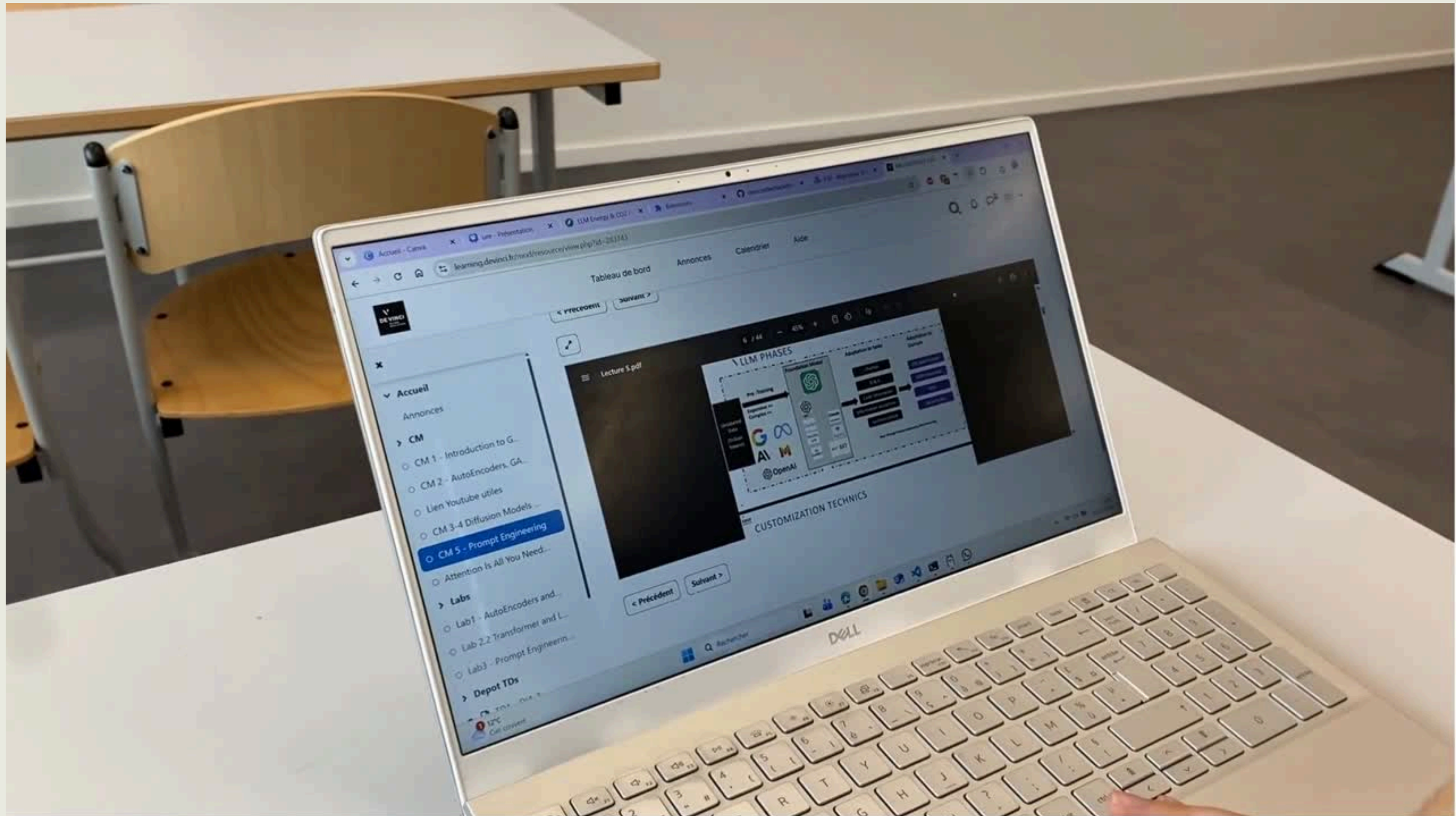
Energy Mix
Carbon Footprint

Send results

CHALLENGES FACED

CATEGORY	CHALLENGE	DESCRIPTION
 ACCESS & SETUP	CONNECTION ISSUES	SOME TEAM MEMBERS COULDN'T ACCESS THE WATSONX APPLICATION.
 COLLABORATION	USER ADDITION PROBLEM	DIFFICULTIES ADDING USERS TO THE SHARED PROJECT WORKSPACE.
 LEARNING CURVE	TOOL FAMILIARITY	TIME NEEDED TO UNDERSTAND AND HANDLE THE WATSONX AND DATA REFINERY TOOLS.
 INTERFACE DESIGN	BUILDING THE INTERFACE	HARD TO SET UP A FUNCTIONAL AND AESTHETIC UI FOR THE SIMULATION.
 INTEGRATION	FRONT/BACK-END LINK	CHALLENGES CONNECTING BOTH LAYERS AND ENSURING DATA FLOW CONSISTENCY.

DEMO



CONCLUSION & FUTURE IMPROVEMENTS

- Propose an optimized query
- Monitor daily consumption (database)
- Improve the user interface
- Extend to other non-open-source models